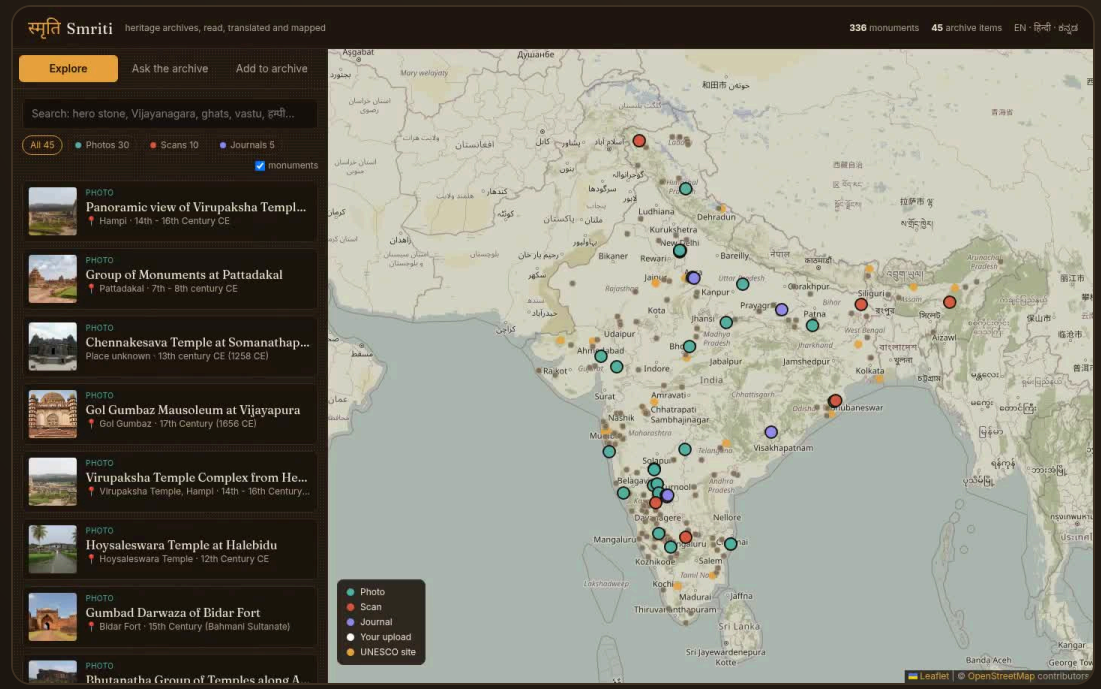
स्मृति · ಸ್ಮೃತಿ

Smriti

Heritage archives, read, translated and mapped.

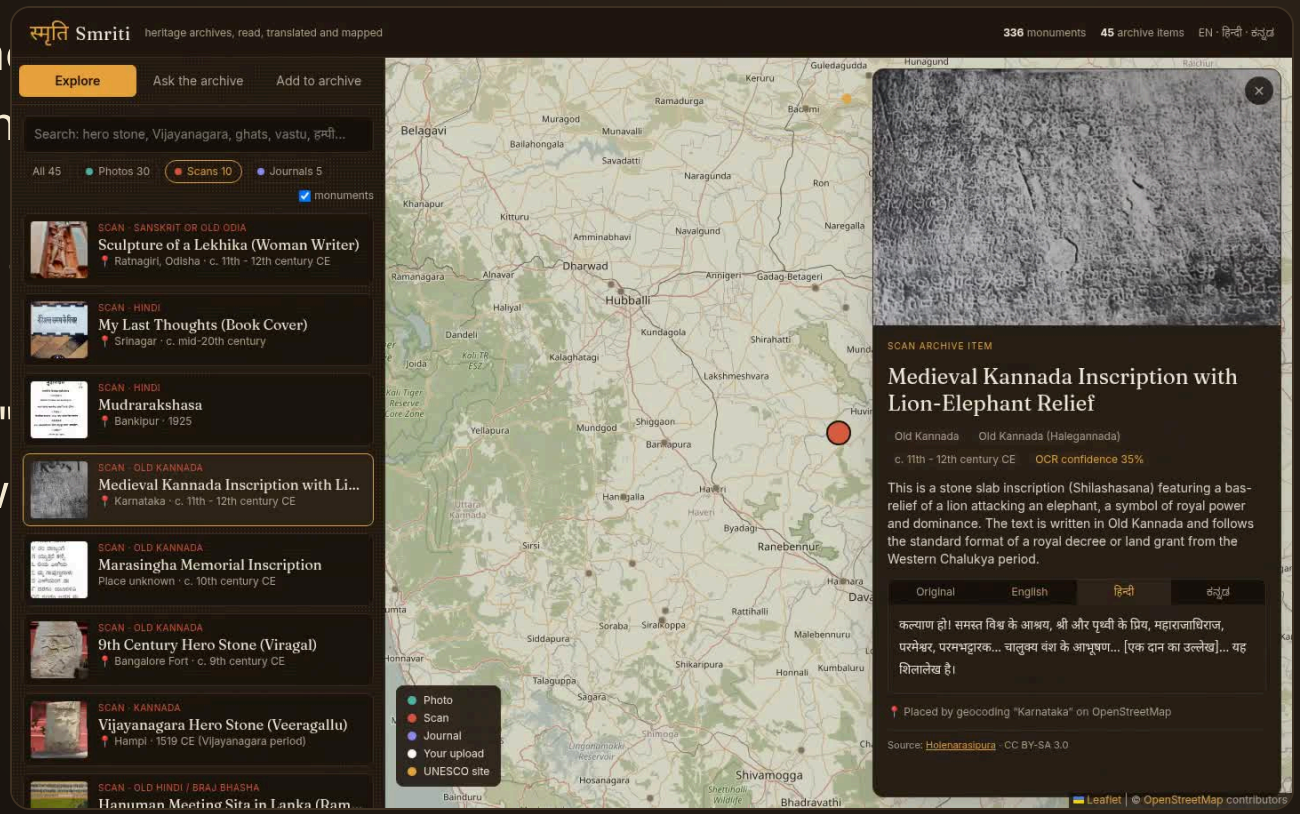
HackCult '26 · Track 2 Culture & Digital Heritage · HC26-CULT01

smriti-ebon-eight.vercel.app · github.com/amantebriwal4321/smriti

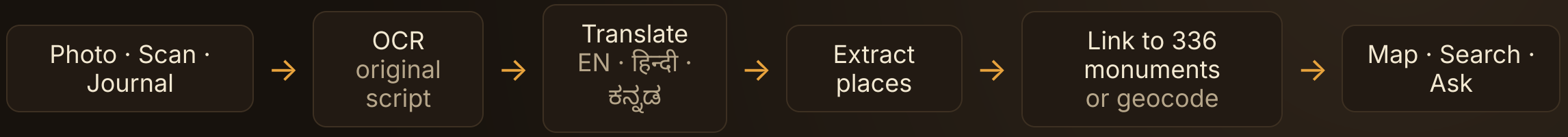


The problem

- Heritage lives in scattered archives: monum
  photos, Devanagari manuscripts, worn Kann
  stones, colonial-era travel journals.
- They are hard to read, rarely translated, not
  places, and impossible to query.
- A student, tourist or researcher cannot ask "
  was written about Hampi?" and get an answ

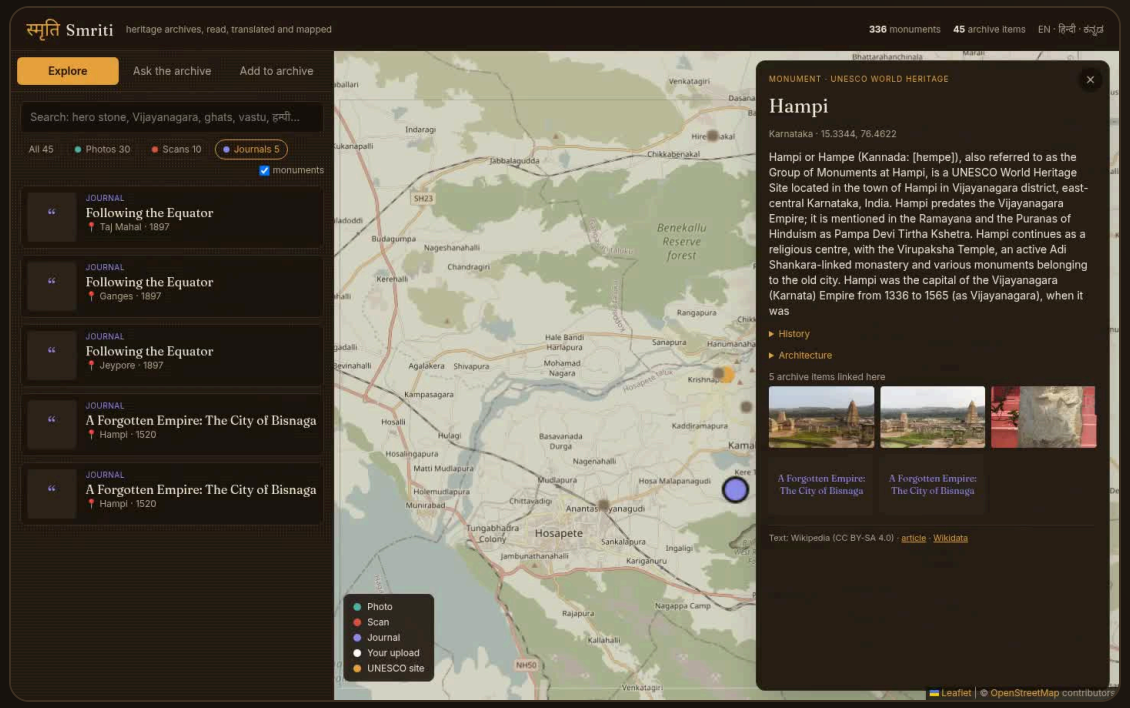


What Smriti does

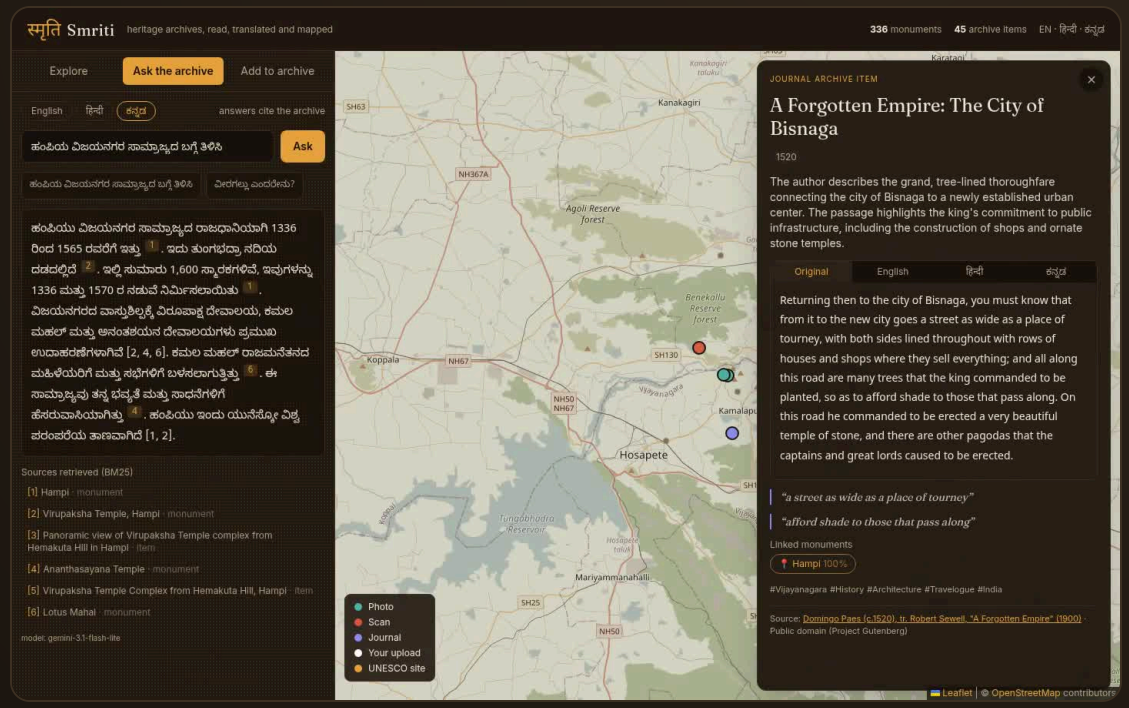


- Worn text is marked [...] with a confidence score, never invented.
- Every item keeps its author, license and source link.
- Ask in English, Hindi or Kannada, get a cited answer, click a citation to fly to the source.
- Anyone can drop in a new scan and watch it land on the map in seconds.

Demo



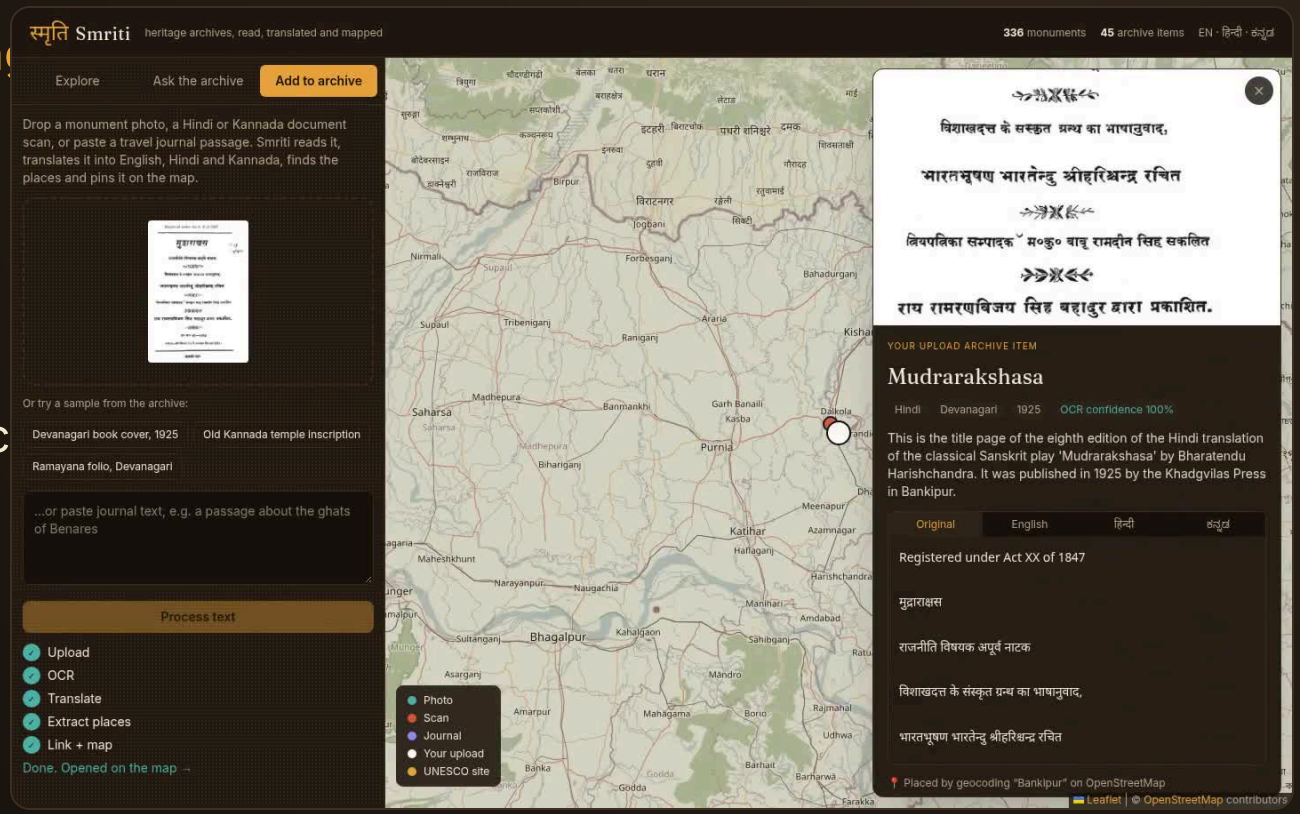
Domingo Paes, c. 1520, linked to Hampi with every related photo and scan



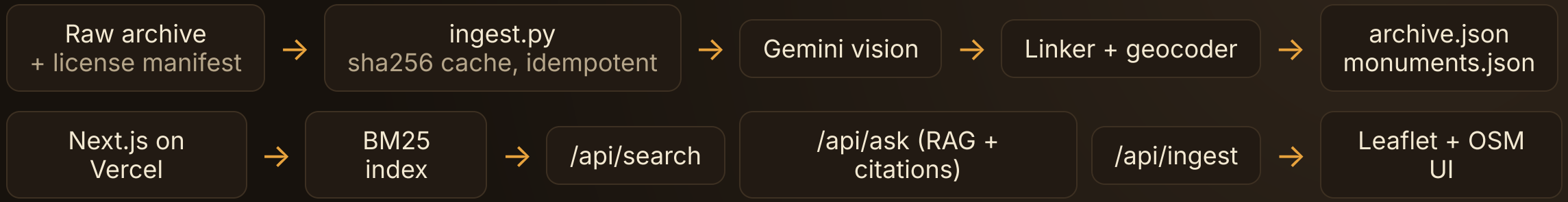
Question in Kannada, answer in Kannada, with clickable citations

Live ingest

- Image resized in the browser, sent to /api/ing
- Gemini vision returns transcription, three translations, era, places and tags as JSON.
- Gazetteer linker: fuzzy match + aliases (Vijayanagar, Bisnaga → Hampi).
- No match? OpenStreetMap Nominatim geoc
- Pin appears on the map, detail opens.



Architecture



- Stateless API routes, horizontally scalable on serverless functions.
- Model rotation with timeouts, so one rate-limited model does not take the app down.
- Pipeline stages map 1:1 to queue jobs for large batch uploads.

Stack and data

Frontend	Next.js 16, React 19, Tailwind 4, Leaflet, OpenStreetMap tiles
AI	Gemini vision models for OCR, translation, extraction, cited answers
Retrieval	In-memory BM25 over monuments + archive text, multilingual query rewrite
Geo	Wikidata gazetteer (336 sites), Nominatim fallback
Seed archive	30 photos + 10 scans from Wikimedia Commons (CC0, CC BY-SA, PD), 5 public-domain journals (Twain 1897, Paes via Sewell 1900)
Hosting	Vercel, keys in env only, MIT licensed repo

Limits and roadmap

Honest limits

- Old Kannada stones are often only partly readable.
- Some manuscripts name no place and stay off the map.
- Live uploads live in the browser session for now.

Next

- Postgres + pgvector, embeddings, persistent uploads.
- Queue workers for full state archives.
- Crowd corrections on OCR, scholar review.
- Timeline view by era, ASI and museum feeds.

Memory, made searchable.

smriti-ebon-eight.vercel.app

github.com/amantebriwal4321/smriti